

KINETIC THEORY OF GASES

Important formulas and concepts directly from NCERT-

❖ Behaviour of gases-

➤ For a sample of gas,

$$P_1 V_1 / N_1 T_1 = P_2 V_2 / N_2 T_2 = \text{constant} = k_B$$

Here, k_B = boltzman's constant = 1.38×10^{-23} J/K

P = pressure

V = volume

T = temperature

N = no. of molecules

➤ $PV = Nk_B T$

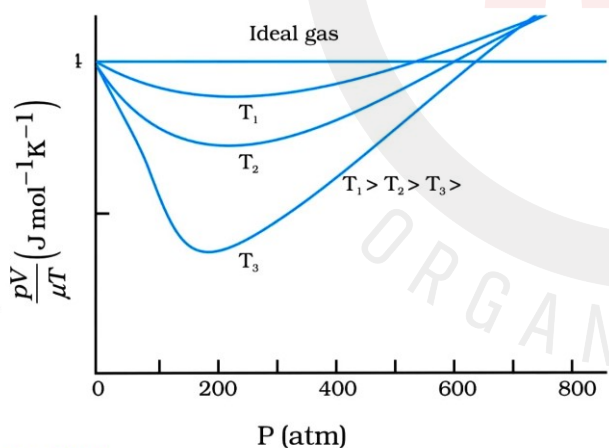


Fig.12.1 Real gases approach ideal gas behaviour at low pressures and high temperatures.

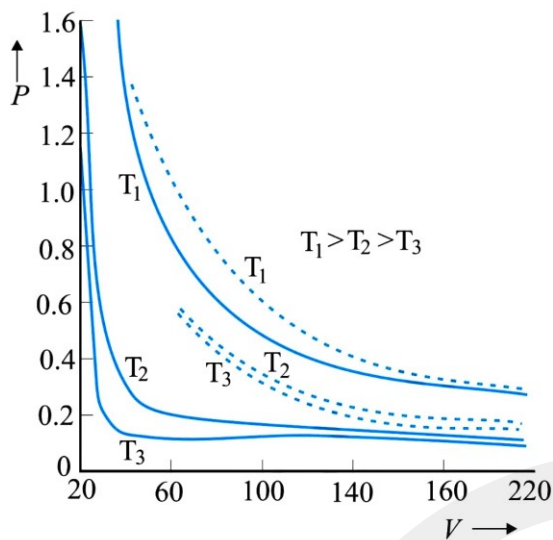


Fig. 12.2 Experimental P - V curves (solid lines) for steam at three temperatures compared with Boyle's law (dotted lines). P is in units of 22 atm and V in units of 0.09 litres.

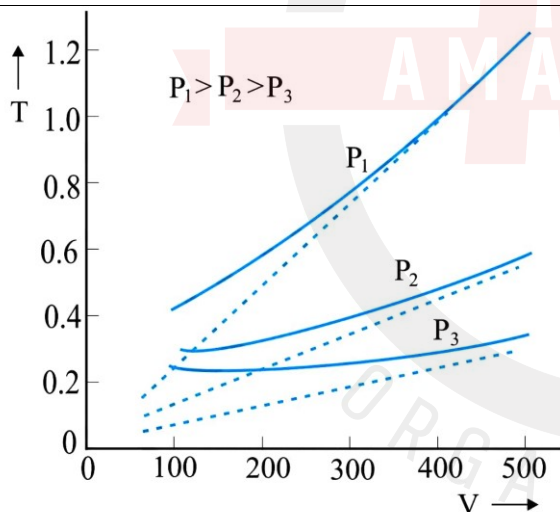


Fig. 12.3 Experimental T - V curves (solid lines) for CO_2 at three pressures compared with Charles' law (dotted lines). T is in units of 300 K and V in units of 0.13 litres.

❖ Pressure of an Ideal gas-

➤ $P = (1/3) nmv^2$

Here, n = no. of gas molecules per unit volume

m = mass of one molecule

v^2 = mean of squared speed

❖ Kinetic interpretation of temperature-

➤ Internal energy E of gas is purely kinetic.

$$PV = (2/3) E$$

➤ Kinetic interpretation,

$$E = (3/2) Nk_B T$$

Here, N = total no. of molecules ($N = nV$)

❖ Specific heat capacity-

➤ $C_P - C_V = R$

➤ $\gamma = C_P/C_V$

[1] monoatomic gas-

○ $C_P = 5R/2$, $C_V = 3R/2$

○ $\gamma = 5/3$

[2] diatomic gas-

○ $C_P = 7R/2$, $C_V = 5R/2$

○ $\gamma = 7/5$

➤ If it has an additional vibrational mode,

$$\gamma = 9/7$$

[3] polyatomic gas-

○ $C_P = (4 + f) R$, $C_V = (3 + f) R$

SUMMARY

- The ideal gas equation connecting pressure (P), volume (V) and absolute temperature (T) is

$$PV = \mu RT = k_B NT$$

where μ is the number of moles and N is the number of molecules. R and k_B are universal constants.

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}, \quad k_B = \frac{R}{N_A} = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

Real gases satisfy the ideal gas equation only approximately, more so at low pressures and high temperatures.

- Kinetic theory of an ideal gas gives the relation

$$P = \frac{1}{3} n m \overline{v^2}$$

where n is number density of molecules, m the mass of the molecule and $\overline{v^2}$ is the mean of squared speed. Combined with the ideal gas equation it yields a kinetic interpretation of temperature.

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T, \quad v_{rms} = (\overline{v^2})^{1/2} = \sqrt{\frac{3k_B T}{m}}$$

This tells us that the temperature of a gas is a measure of the average kinetic energy of a molecule, *independent of the nature of the gas or molecule*. In a mixture of gases at a fixed temperature the heavier molecule has the lower average speed.

- The translational kinetic energy

$$E = \frac{3}{2} k_B NT.$$

This leads to a relation

$$PV = \frac{2}{3} E$$

- The law of equipartition of energy states that if a system is in equilibrium at absolute temperature T , the total energy is distributed equally in different energy modes of absorption, the energy in each mode being equal to $\frac{1}{2} k_B T$. Each translational and rotational degree of freedom corresponds to one energy mode of absorption and has energy $\frac{1}{2} k_B T$. Each vibrational frequency has two modes of energy (kinetic and potential) with corresponding energy equal to

$$2 \times \frac{1}{2} k_B T = k_B T.$$

- Using the law of equipartition of energy, the molar specific heats of gases can be determined and the values are in agreement with the experimental values of specific heats of several gases. The agreement can be improved by including vibrational modes of motion.
- The mean free path l is the average distance covered by a molecule between two successive collisions :

$$l = \frac{1}{\sqrt{2} n \pi d^2}$$

where n is the number density and d the diameter of the molecule.

POINTS TO PONDER

1. Pressure of a fluid is not only exerted on the wall. Pressure exists everywhere in a fluid. Any layer of gas inside the volume of a container is in equilibrium because the pressure is the same on both sides of the layer.
2. We should not have an exaggerated idea of the intermolecular distance in a gas. At ordinary pressures and temperatures, this is only 10 times or so the interatomic distance in solids and liquids. What is different is the mean free path which in a gas is 100 times the interatomic distance and 1000 times the size of the molecule.
3. The law of equipartition of energy is stated thus: the energy for each degree of freedom in thermal equilibrium is $\frac{1}{2} k_B T$. Each quadratic term in the total energy expression of a molecule is to be counted as a degree of freedom. Thus, each vibrational mode gives 2 (not 1) degrees of freedom (kinetic and potential energy modes), corresponding to the energy $2 \times \frac{1}{2} k_B T = k_B T$.
4. Molecules of air in a room do not all fall and settle on the ground (due to gravity) because of their high speeds and incessant collisions. In equilibrium, there is a very slight increase in density at lower heights (like in the atmosphere). The effect is small since the potential energy (mgh) for ordinary heights is much less than the average kinetic energy $\frac{1}{2} mv^2$ of the molecules.
5. $\langle v^2 \rangle$ is not always equal to $(\langle v \rangle)^2$. The average of a squared quantity is not necessarily the square of the average. Can you find examples for this statement.

Dr. Aman Tilak
MBBS, AIIMS Delhi